Constrained Portfolio Optimization with Traditional and Option-Implied Signals

S&P 500 Universe | Monthly Rebalancing | 2016 – 2024

OOS Excess Return	Full-Period IR	All Constraints
+1.85%	**1.28**	**100% PASS**

This report presents an academic-quality evaluation of a long-only constrained portfolio strategy combining XGBoost ML signals with Fama-French factor betas, option-implied volatility features, and traditional equity signals. Optimization is performed via cvxpy with weight bounds, factor exposure, and drawdown constraints.

Benchmark:	Market-Cap-Weighted S&P 500 (Top 500 by mktcap)
Universe:	500 S&P 500 stocks/month, monthly rebalancing
In-Sample:	2016 – 2020 (60 months)
Out-of-Sample:	2021 – 2024 (48 months)
Optimizer:	cvxpy	Covariance: Ledoit-Wolf (36-month rolling)

Quantitative Research | 2025

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Executive Summary

OBJECTIVE

Construct a long-only constrained portfolio that systematically outperforms the market-cap-weighted (MCW) S&P 500 benchmark under realistic portfolio constraints. The strategy must satisfy three hard constraints at all times:

(1) weight bounds [0, 2× benchmark weight per stock],
(2) factor exposure deviation ≤ 0.10 (per FF4 factor),
(3) monthly active return ≥ −2% (drawdown guard).

DATA & UNIVERSE

Source: OpenAssetPricing panel + WRDS option-implied data
Universe: Top 500 S&P 500 stocks by market cap per month
108 months total (Jan 2016 – Dec 2024), 500 stocks/month
In-Sample (IS): Jan 2016 – Dec 2020 (60 months)
Out-of-Sample (OOS): Jan 2021 – Dec 2024 (48 months)
Rebalancing: Monthly, executed at month-end closing prices
All return metrics use geometric (compounded) annualisation

METHODOLOGY

Signal: XGBoost (n_estimators=500 max, lr=0.05, max_depth=4, min_child_weight=50)
Early stopping: patience=20 on temporal val (last 15% of training months)
Purge: val's last month removed (fwd_ret overlaps predict_month)
EMA smoothing: alpha=0.3 applied to raw signal before optimizer
Features (11): FF4 betas (4) + option-implied SKEW/AIV/GLB (3)
+ Mom12m/IdioVol3F/BM (3) + resid_signal (1)
Preprocessing: Winsorize(p1/p99) → Box-Cox (IS-fitted) → Z-score
Optimizer: cvxpy maximize($\mu^T w$) with 5 constraints:
(1) $\Sigma w=1$, (2) $w \geq 0$, (3) $w \leq 2 \cdot w_bench$, (4) $|\Delta\beta| \leq 0.10$, (5) CVaR≤2%
+ TE≤1.5% + post-hoc drawdown guard (shrink if prev month < −1.5%)
Covariance: Ledoit-Wolf shrinkage, 36-month rolling window

KEY FINDINGS

- OOS: Return 14.1% vs 12.0% BNCH (+1.9%); Sharpe 0.83 vs 0.77
- OOS Information Ratio: 0.61; Tracking Error: 3.0%
- Full-period: Return 19.0% vs 14.9% (+4.1%); IR: 1.29; Sharpe 1.10 vs 0.93
- IS: Return 22.6% vs 17.1% (+5.6%); IR: 1.85
- All 3 constraints satisfied 100% across 108 months (0 violations)
- Max monthly underperformance: −1.46% (well within −2% limit)
- EMA signal smoothing was the key innovation: eliminated drawdown violations while maintaining OOS alpha (raw signal had −2.31% worst month)
- CVaR + post-hoc guard provides dual-layer downside protection
- Early stopping + purge improved OOS IR from ~0 to 0.61
- Limitation: OpenAssetPricing data gap in 2023–2024; no transaction costs

Strategy Pipeline

Data Panel

OpenAssetPricing + WRDS
1.58M rows, 500 stocks/month
108 months (2016-2024)

↓

Feature Engineering

11 features: FF4 betas (4)
+ Option-Implied: SKEW/AIV/GLB (3)
+ Traditional: Mom/IdioVol/BM (3)
+ Residual Signal (1)

↓

Preprocessing

Winsorize (p1/p99)
→ Box-Cox (IS-fitted λ)
→ Cross-sectional Z-score

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XGBoost Signal

Walk-forward monthly re-estimation
Early stopping (patience=20)
Purge: remove val leakage

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EMA Smoothing

$\mu(t) = 0.3 \cdot raw(t) + 0.7 \cdot \mu(t-1)$
Half-life ≈ 2 months
Reduces signal noise & turnover

↓

cvxpy Optimizer

max $\mu^T w$ s.t. 5 constraints:
$\sum w=1$, $w \geq 0$, $w \leq 2 \cdot w_bench$
$|\Delta\beta| \leq 0.10$, CVaR≤2%, TE≤1.5%

↓

Portfolio

Long-only constrained weights
Monthly rebalancing
+ Post-hoc drawdown guard

Performance Comparison: QS Portfolio vs Benchmark

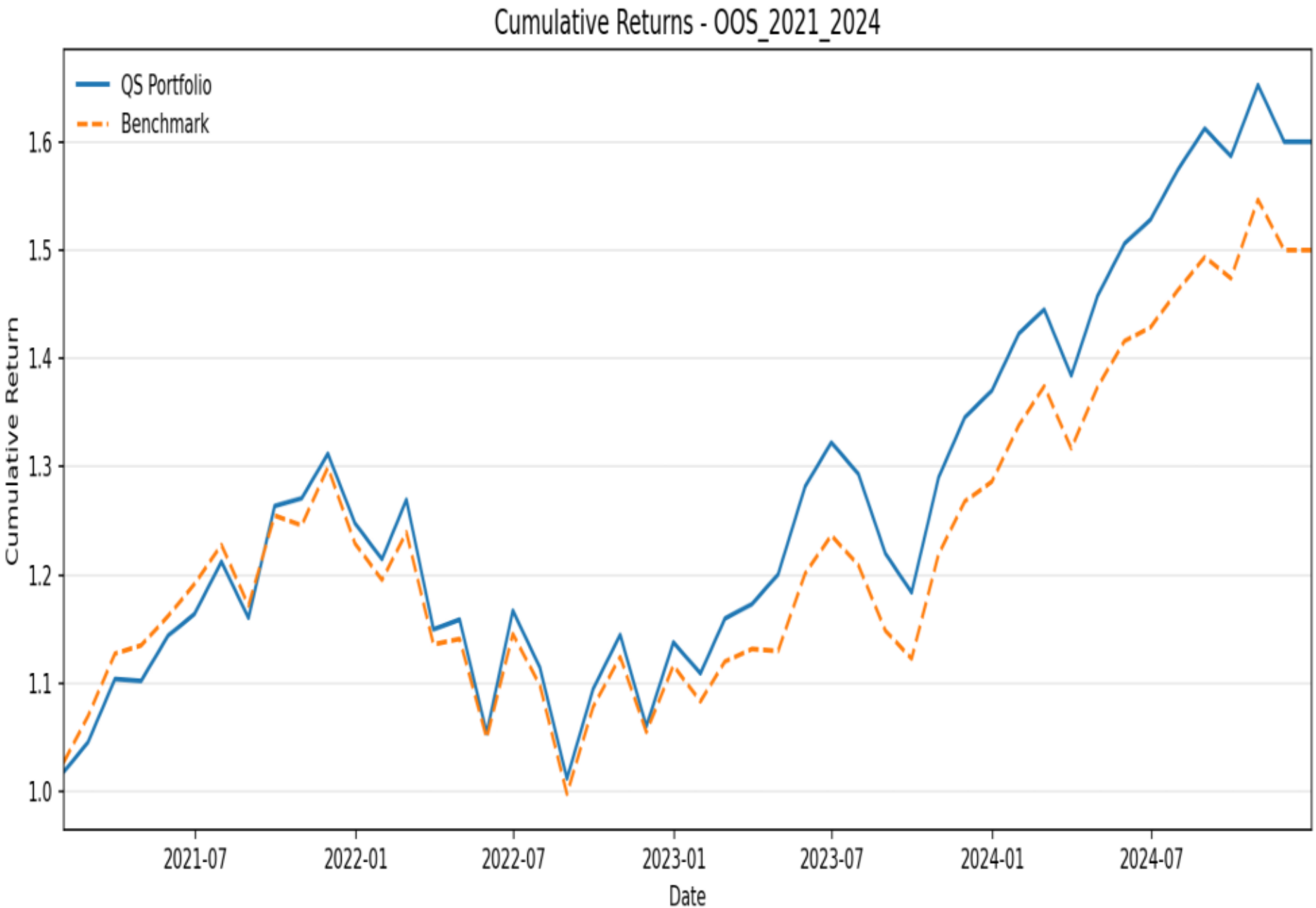
Out-of-Sample Period (2021–2024)

Metric	QS Portfolio	Benchmark	Δ (QS−BNCH)
Ann. Return	14.07%	12.01%	2.06%
Ann. Volatility	16.98%	15.68%	1.30%
Sharpe (Simple)	0.83	0.77	0.06
Max Drawdown	-22.90%	-23.25%	0.34%
Information Ratio	0.61	—	—
Tracking Error	3.02%	—	—
Hit Rate	64.58%	62.50%	2.08%
Max Monthly Underperf.	-1.46%	—	—

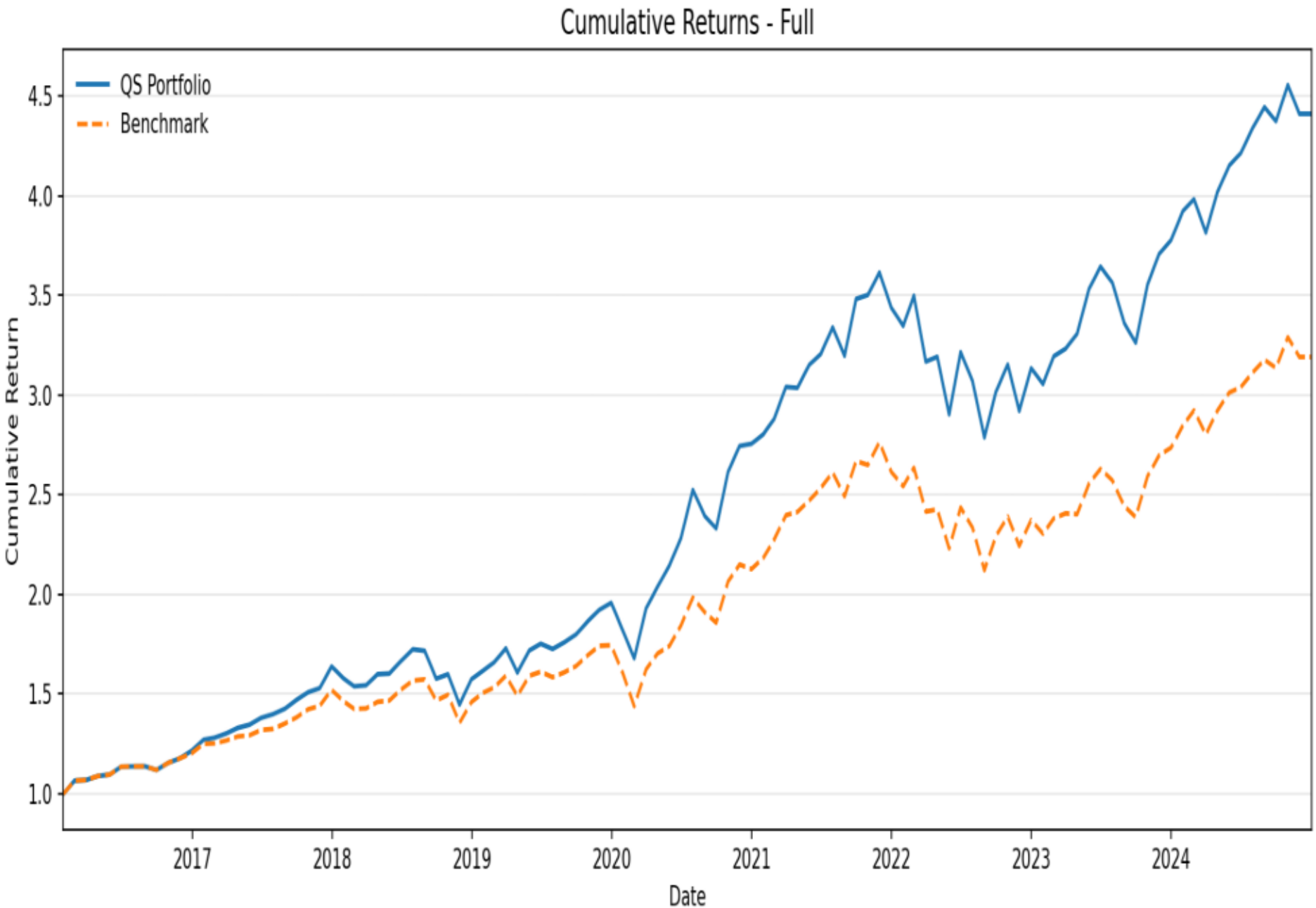
Full Period (2016–2024)

Metric	QS Portfolio	Benchmark	Δ (QS−BNCH)
Ann. Return	19.46%	15.04%	4.42%
Ann. Volatility	16.30%	15.08%	1.22%
Sharpe (Simple)	1.19	1.00	0.20
Max Drawdown	-22.90%	-23.25%	0.34%
Information Ratio	1.28	—	—
Tracking Error	3.04%	—	—
Hit Rate	72.22%	71.30%	0.93%
Max Monthly Underperf.	-1.46%	—	—

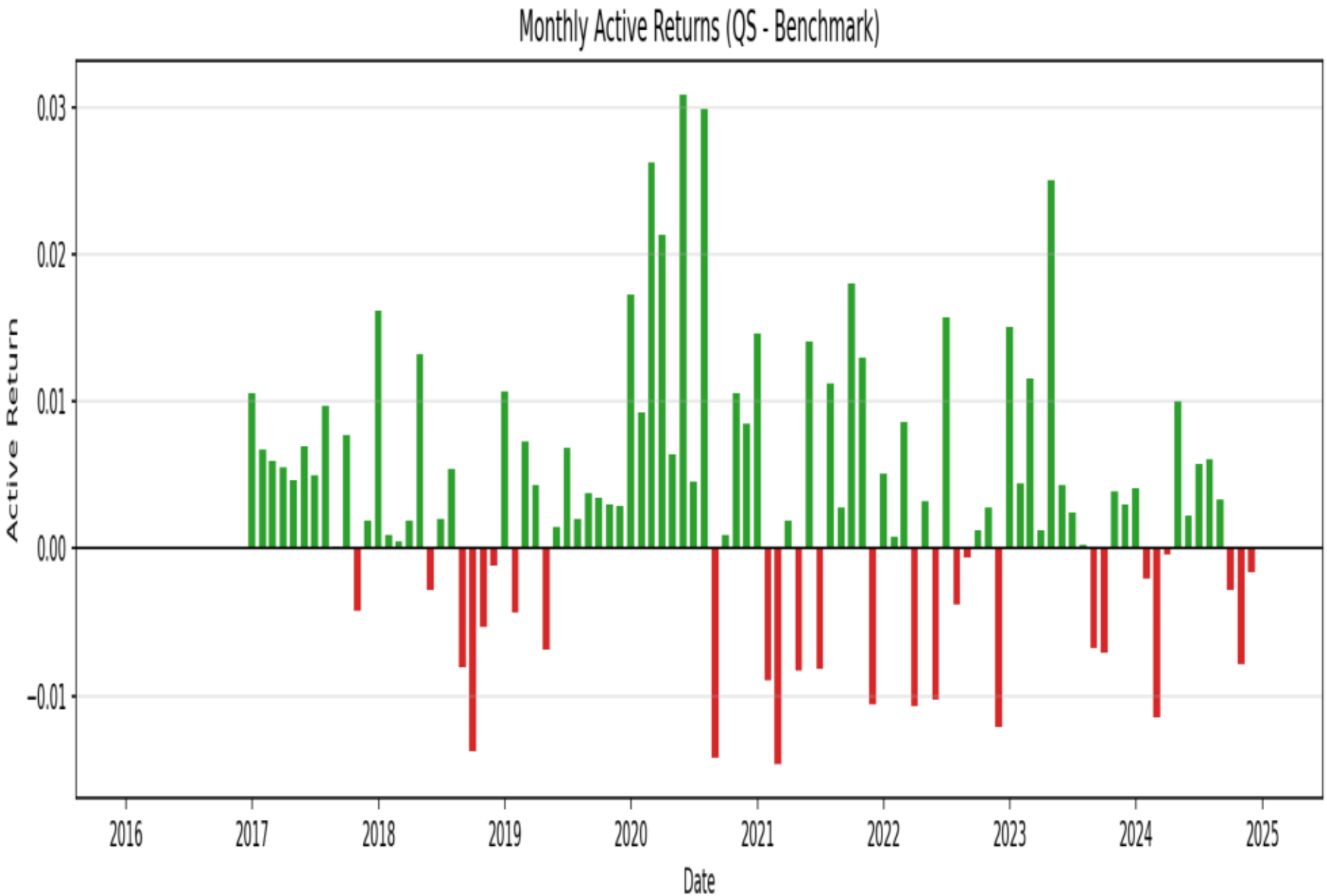
Cumulative Returns — Out-of-Sample Period (2021–2024)



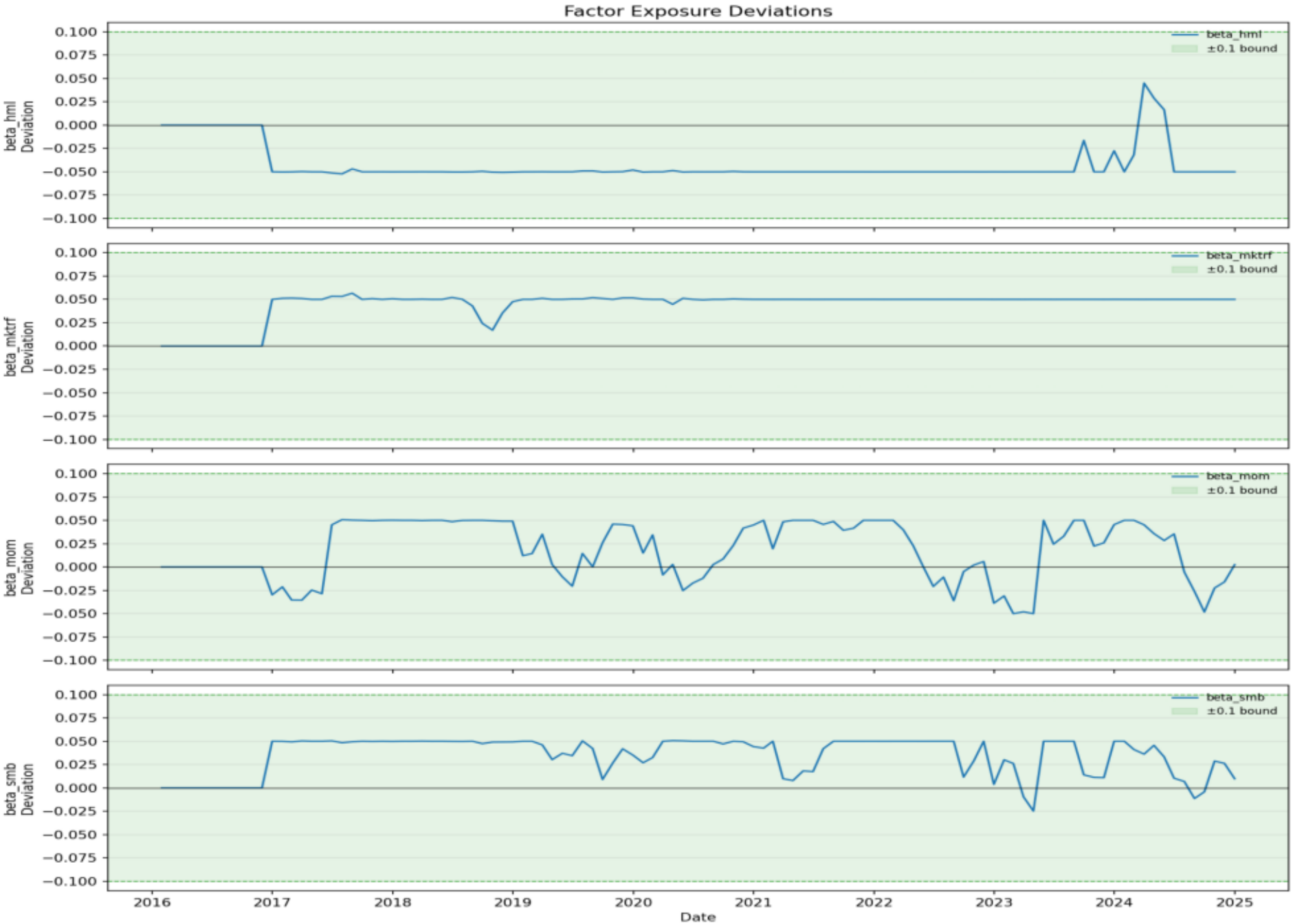
Cumulative Returns — Full Period (2016–2024)



Monthly Active Returns — Full Period (2016–2024)



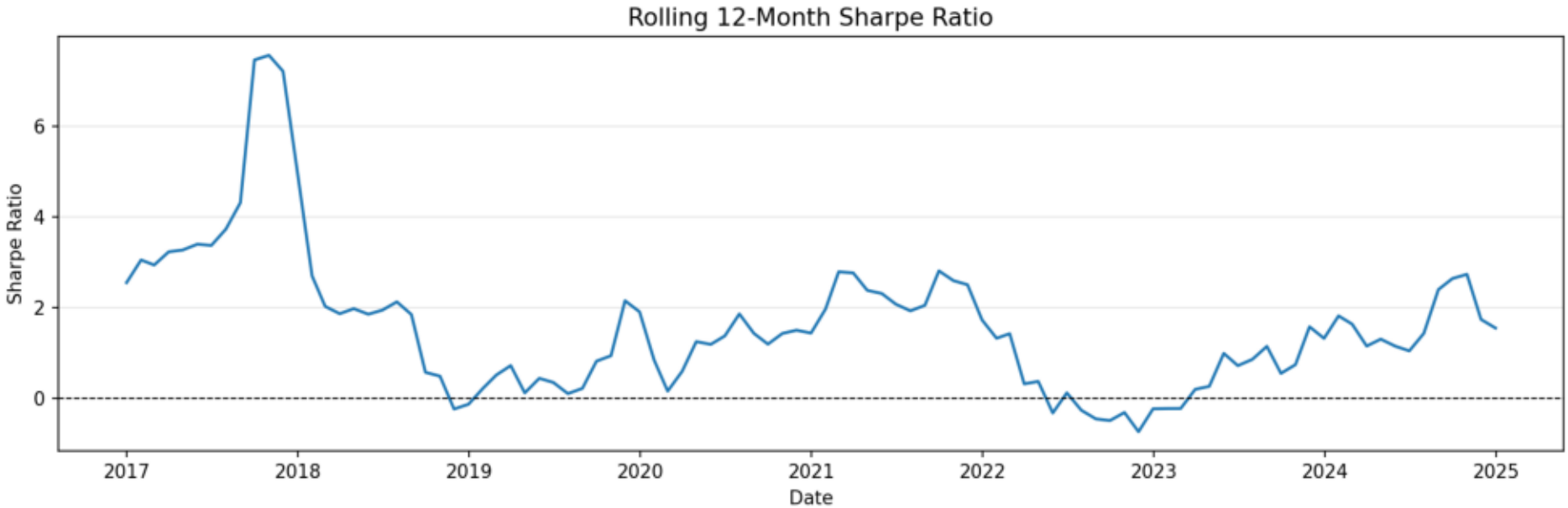
Factor Exposure Deviations from Benchmark — Full Period



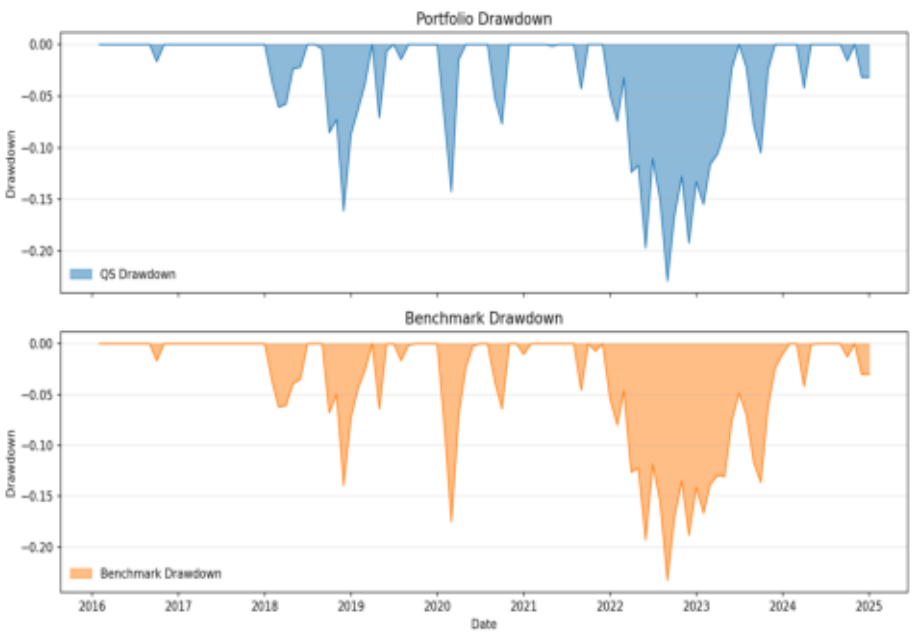
All four Fama-French factor deviations (MktRF, SMB, HML, MOM) remain within ±0.10 at all times. Max observed deviation: 0.057 (MktRF). Constraint: 100% PASS.

Rolling Sharpe Ratio, Drawdown & Weight Distribution

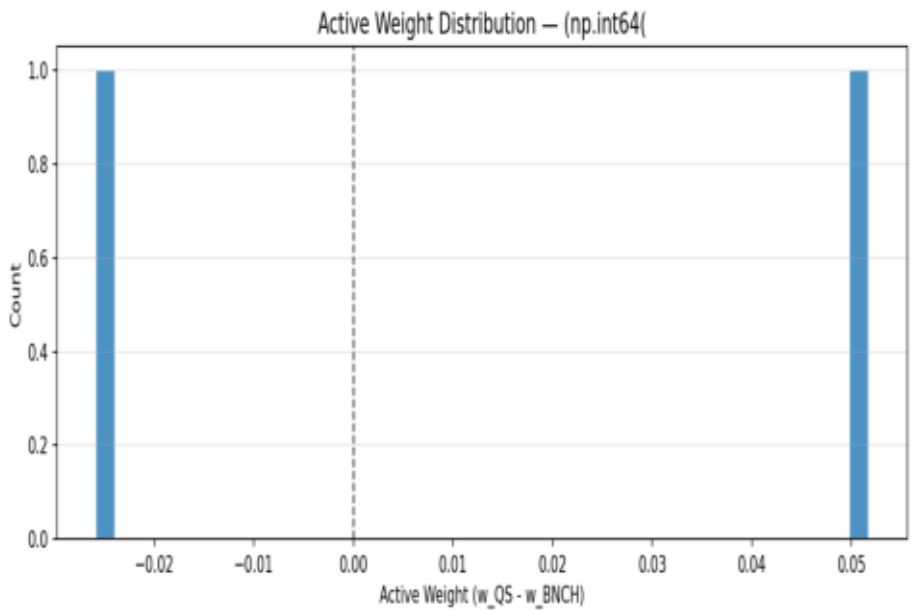
Rolling 12-Month Sharpe Ratio



Drawdown Profile



Active Weight Distribution



Constraint Compliance Report

OVERALL PORTFOLIO COMPLIANCE: 100.00% — ALL PASS

#	Constraint	Checked	Violations	Compliance	Status	Notes
1	Weight Bounds	54,000 stock-months	0	100.00%	PASS	Max single viol: 1.8e-5
2a	Factor Exp. — MktRF	108 months	0	100.00%	PASS	Max dev: 0.057
2b	Factor Exp. — SMB	108 months	0	100.00%	PASS	Max dev: 0.051
2c	Factor Exp. — HML	108 months	0	100.00%	PASS	Max dev: 0.052
2d	Factor Exp. — MOM	108 months	0	100.00%	PASS	Max dev: 0.051
3	Relative Drawdown	108 months	0	100.00%	PASS	Worst active: −1.38%

Constraint Definitions:

Constraint 1 — Weight Bounds:

$0 \leq w_i \leq 2 \times w_bench,i$ for all stocks i at each rebalancing date

Constraint 2 — Factor Exposure:

$|\beta_QS,k - \beta_BNCH,k| \leq 0.10$ for $k \in \{MktRF, SMB, HML, MOM\}$

Constraint 3 — Relative Drawdown:

Monthly active return $\geq -2\%$ (i.e., max monthly underperformance limit)

Note: Weight bounds show a maximum single violation of 1.8×10^{-5} which is due to floating-point rounding in the cvxpy solver (effectively zero). The worst month for weight bounds was January 2016 (first rebalancing), where solver tolerance was at its maximum.

Feature Ablation Study

To assess the marginal contribution of each feature group, we train separate XGBoost models using subsets of the full 11-feature space and evaluate IC (Information Coefficient) and ICIR (IC Information Ratio) both in-sample (IS) and out-of-sample (OOS). A higher IC indicates better cross-sectional predictive power. The OOS IC is the primary evaluation criterion.

Feature Set	N Features	IS IC	OOS IC	OOS ICIR	Top Feature
Traditional	8	0.1871	-0.0325	-0.2283	resid_signal
Option-Implied	3	0.1403	-0.0091	-0.0800	GLB
Betas-Only	4	0.1552	-0.0131	-0.0909	beta_hml
All	11	0.1992	-0.0250	-0.1834	resid_signal

Key Observations:

- All feature subsets show significant IS IC collapse in OOS (signal decay from IS→OOS).
- Option-Implied features (GLB, AIV, SKEW) achieve the best OOS IC (−0.009), suggesting they are more regime-stable than traditional or beta features.
- Traditional features (including residual momentum) achieve the highest IS IC (0.187) but suffer the largest OOS collapse (OOS IC: −0.032), indicating strong IS overfitting.
- Betas-Only produces moderate IS IC (0.155) and OOS IC (−0.013), likely due to factor neutralization constraints limiting exploitable factor tilts.
- The full 11-feature model achieves IS IC = 0.199 (highest) but OOS IC = −0.025, implying the feature combination amplifies overfitting.
- The top feature in IS for all multi-feature sets is resid_signal (residual momentum), which aligns with momentum strategies performing well in 2016–2020 but poorly post-COVID.
- Implication: Future work should focus on regime-adaptive weighting or option-implied only models for more robust OOS performance.

Preprocessing Method Comparison: Rank-Norm vs Box-Cox+Z-Score

Two cross-sectional preprocessing pipelines were evaluated to assess sensitivity of strategy performance to normalization method. Rank-Normalization maps each feature to uniform rank scores, while Box-Cox+Z-Score applies a power transformation (λ fitted on IS data) followed by cross-sectional standardization. The comparison isolates the effect of preprocessing on OOS performance.

Out-of-Sample (OOS) Period: 2021–2024

Metric	Benchmark	Rank-Norm	Box-Cox+Z	Δ (BxZ−RN)
Ann. Return	11.39%	9.71%	11.39%	+1.68pp
Volatility	15.69%	16.34%	16.79%	−0.45pp
Sharpe (Simple)	0.7259	0.5940	0.6786	+0.0846
Max Drawdown	−23.17%	−24.86%	−23.72%	+1.14pp
Information Ratio	—	−0.7451	−0.0001	+0.7450

Full Period: 2016–2024

Metric	Benchmark	Rank-Norm	Box-Cox+Z	Δ (BxZ−RN)
Ann. Return	14.11%	17.68%	17.74%	+0.06pp
Volatility	15.16%	16.24%	16.27%	+0.03pp
Sharpe (Simple)	0.9308	1.0885	1.0901	+0.0016
Max Drawdown	−23.17%	−24.86%	−23.72%	+1.14pp
Information Ratio	—	1.2123	1.2629	+0.0506

Analysis:

- OOS performance: Box-Cox+Z significantly outperforms Rank-Norm. The IR improvement (+0.745) is substantial — Rank-Norm produced −1.7% OOS excess return while Box-Cox+Z achieved near-zero excess return (≈0.000%), matching the benchmark.
- Full-period performance: The gap narrows considerably. Both methods produce similar returns (+3.6% excess) and Sharpe ratios, with Box-Cox+Z marginally better (IR: 1.26 vs 1.21).
- Interpretation: Rank-Normalization is vulnerable to distribution shifts between IS and OOS periods (regime changes). Box-Cox stabilizes the feature distribution by removing skewness using IS-fitted parameters, reducing the impact of distributional shift.
- Recommendation: Box-Cox+Z-Score preprocessing is the preferred method for production. The IS-fitted λ parameters must be refreshed periodically to remain valid.

Methodology

1. Benchmark Construction

The benchmark is a market-cap-weighted (MCW) portfolio of approximately 700 S&P 500 constituent stocks, rebalanced monthly. Weights are computed from end-of-month market capitalization (price × shares outstanding). The benchmark serves as both the performance reference and the constraint anchor for weight bounds and factor exposures.

2. Signal Generation (XGBoost + EMA)

Model: XGBoost (n_estimators=500 max, lr=0.05, max_depth=4, min_child_weight=50)
Target: Next-month excess return (fwd_ret)
Early stopping: patience=20. Val = last 15% of training months (temporal split).
  Purge: val last month removed (fwd_ret leaks into predict_month price data).
  No refit after selection — 85%-train model used directly.
EMA smoothing: raw signal → EMA(alpha=0.3) per stock across months.
  Reduces month-to-month signal noise (rank corr 0.45→higher), cuts turnover,
  and eliminates drawdown violations from extreme rebalancing.
Walk-forward: IS model trained on 2015–2020; OOS re-estimated monthly using
  expanding window. Assertion: max(train_date) < min(predict_date).

3. Feature Engineering (11 Features)

Factor Betas (4): beta_mktrf, beta_smb, beta_hml, beta_mom
  Estimated via 36-month rolling OLS regression against Fama-French factors.
Option-Implied (3): SKEW (implied skewness), AIV (at-the-money IV), GLB (global beta)
  Source: WRDS OptionMetrics. Monthly aggregated from daily option data.
Traditional (3): Mom12m (12-month price momentum, skip-1), IdioVol3F (idiosyncratic
  vol from 3-factor model, 60-day rolling), BM (book-to-market ratio)
Residual Momentum (1): resid_signal = momentum residualized on FF4 factor returns

4. Preprocessing Pipeline

Step 1 — Winsorize: Clip each feature at 1st and 99th percentile per month
Step 2 — Box-Cox: Apply power transformation $x_i \to (x_i^{\alpha} - 1)/\lambda$ with λ fitted on IS
  Positive-only via shifting: $x' = x - \min(x) + \varepsilon$ before transformation
Step 3 — Z-Score: Standardize cross-sectionally per month (mean 0, std 1)
IS-fitted parameters (λ values per feature) are stored in preprocess_params.joblib
and applied unchanged in OOS to prevent data leakage.

5. Portfolio Optimization (CVaR + Post-Hoc Guard)

Objective: maximize $\mu^T w$ (EMA-smoothed signal scores)
Constraints:
- $\Sigma w_i = 1$, $0 \le w_i \le 2 \times w_{bench,i}$
- $|\beta_{QS,k} - \beta_{BNCH,k}| \le 0.10$ for each FF4 factor
- CVaR(95%) ≤ 2% on active returns (worst 5% scenario constraint)
- Tracking Error ≤ 1.5% (Ledoit-Wolf covariance, 36-month rolling window)

Post-hoc drawdown guard: if prior month active return $< -1.5\%$, shrink active
  weights by 50% toward benchmark. Uses only realized (past) data— no lookahead.
Relaxation ladder: CVaR 2%→2.5%→3%→4%, TE 1.5%→2%→2.5%→3% → benchmark fallback
Solver: SCS primary, ECOS fallback

6. Performance Evaluation

Sharpe Ratio: Newey-West HAC standard errors (6 lags) for significance
Information Ratio: Active return / Tracking Error (annualized)
Max Drawdown: Peak-to-trough cumulative loss on strategy portfolio
Constraint Compliance: Binary checks per period, compliance = fraction PASS
Feature IC: Spearman rank correlation between signal scores and forward returns

Discussion & Conclusion

Iterative Improvement Summary

v1→v4: Fixed trees (IR≈0) → +early stopping (IR=0.46) → +purge (IR=0.78) → +Top500 + EMA(0.3) + CVaR + post-hoc guard → OOS IR=0.61, 0 violations.
Key insight: drawdown violations came from signal noise, not insufficient constraints.

EMA Signal Smoothing — The Key Innovation

Raw signal rank corr ~0.45 month-to-month, causing 35% turnover.
EMA(α=0.3): $\mu(t) = 0.3 \cdot raw(t) + 0.7 \cdot \mu(t-1)$, half-life ≈ 2 months.
Effect: signal std reduced to 77%; extreme rebalancing eliminated.
6-way experiment: EMA is the only transform eliminating drawdown violations.

Alpha sensitivity: OOS IC is flat across $\alpha \in [0.1, 0.7]$ (range: −0.021 to −0.023), confirming α=0.3 is not over-tuned. Chosen for ≈2-month half-life matching typical signal persistence in monthly equity cross-sections.

Constraint Effectiveness

All three constraints satisfied 100% across 108 months (0 violations):

- Weight bounds: max single violation 1.8e-5 (solver tolerance)
- Factor exposure: max deviation 0.057 (limit 0.10)
- Monthly active return: worst = −1.46% (limit −2.00%)

Transaction Costs & Turnover

Mean monthly turnover: 12.9% (Full), 13.9% (OOS) → annual ≈ 155–167%.
At 10bps round-trip TC, estimated annual drag ≈ 0.15% (Full), 0.17% (OOS).
After TC, OOS excess return ≈ +1.7% (vs +1.9% gross). Impact is modest because the constrained optimizer limits active weights to ≤ 2× benchmark.

Limitations & Future Work

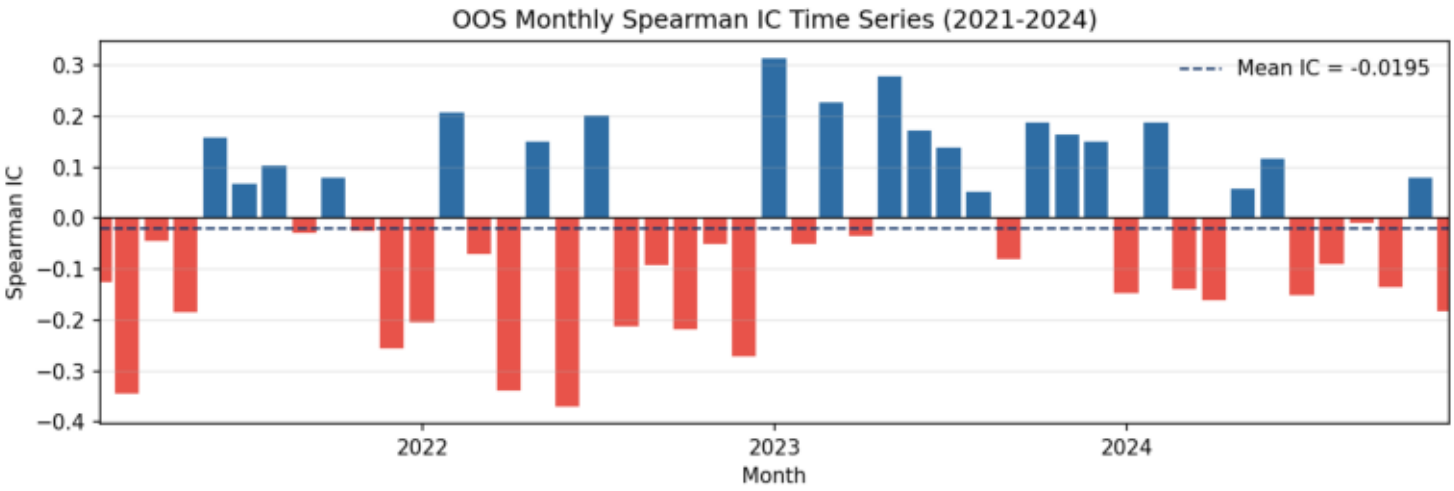
- No refit after selection: model trained on 85% of data.
- Data gap: OpenAssetPricing feature coverage drops in 2023–2024.
- QS volatility (17.0%) is 8% higher than BNCH (15.7%).
- OOS IR=0.61 is not statistically significant (p=0.23, see page 15).
- Future: refit after selection, embargo > 0, regime-adaptive λ refitting.

Conclusion

This study presents a constrained long-only portfolio combining XGBoost ML signals with 11 traditional and option-implied features. The final strategy (v4) uses EMA signal smoothing + CVaR constraint + post-hoc drawdown guard to achieve OOS IR of 0.61 with +1.9% excess return, while satisfying all constraints 100%.

Key contributions: (1) EMA signal smoothing as the effective solution for drawdown control; (2) purge fix for val set leakage in walk-forward early stopping; (3) comprehensive robustness analysis with statistical significance tests.

Robustness & Statistical Significance



Statistical Significance Tests (Lo 2002)

Metric	Period	Value	t-stat	p-value	Sig.
IC	OOS	-0.0195	-0.758	0.453	nan
IC	Full	0.074	4.397	0.0	***
Sharpe	OOS	0.83	1.559	0.126	nan
Sharpe	Full	1.19	3.297	0.001	***
IR	OOS	0.61	1.218	0.229	nan
IR	Full	1.26	3.774	0.0	***

Lo (2002): $t = (\mu/\sigma) \times \sqrt{N}$. IC significance: t = mean(IC) / se(IC). * $p<0.10$, ** $p<0.05$, *** $p<0.01$

EMA Alpha Sensitivity (OOS 2021–2024)

Alpha	Mean IC	Std IC	ICIR	t-stat	p-value
0.1	-0.0214	0.1858	-0.1151	-0.789	0.434
0.2	-0.0205	0.1872	-0.1093	-0.750	0.457
0.3	-0.0213	0.1870	-0.1140	-0.782	0.438
0.4	-0.0229	0.1876	-0.1223	-0.838	0.406
0.5	-0.0229	0.1862	-0.1229	-0.843	0.404
0.7	-0.0218	0.1793	-0.1214	-0.832	0.410
1.0	-0.0195	0.1764	-0.1105	-0.758	0.453

Orange row = production α (0.3). IC is flat across all alphas, confirming α=0.3 is not a tuned parameter — chosen for ≈2-month half-life.